

CSCE 647 IMAGE SYNTHESIS · MODULE 08 · LECTURE NOTES

Path Tracing and Multiple Importance Sampling

The algorithm, and the fifteen lines that make it usable.

MODULE OUTCOMES

- Implement a correct unidirectional path tracer.
- Explain next-event estimation and why it is not optional.
- Derive the MIS estimator and the balance and power heuristics.
- Explain exactly why naive NEE plus BSDF sampling double-counts.
- Diagnose path tracer bugs from the images they produce.

1 The naive path tracer

The algorithm follows directly from Module 02's Neumann series: sample paths of every length with the correct probability and average. The entire integrator fits on one screen.

```
vec3 radiance(Ray r, Scene& s) {
    vec3 L(0), throughput(1);
    for (int depth = 0; ; depth++) {
        Hit h;
        if (!s.intersect(r, h)) { L += throughput * s.env(r.d); break; }
        L += throughput * h.emission();

        auto [wi, f, pdf, flags] = h.bsdf->sample(-r.d, rng.next2D());
        if (pdf <= 0) break;
        throughput *= f * fabs(dot(wi, h.ns)) / pdf;

        if (depth > 3) {
            float q = fmaxf(0.05f, 1.0f - throughput.max());
            if (rng.next() < q) break;
            throughput /= (1.0f - q);
        }
        r = Ray(offset(h.p, h.ng), wi);
    }
    return L;
}
```

This is complete and unbiased. The throughput accumulates the product of $f \cdot \cos / \text{pdf}$ at each vertex — exactly the Monte Carlo weight for that path — and Russian roulette (Module 06) terminates without

truncation bias. Run it on a scene with a large area light and it produces a correct image.

CORRECT, AND UNUSABLE

A path contributes nothing unless it **randomly happens to hit a light source**. The probability of that depends entirely on how much of the hemisphere the light occupies.

A light subtending 0.01 steradians is hit by roughly 1 in 600 cosine-weighted samples. The other 599 cost full price and return zero. For a **point light the probability is exactly zero** and the image is black no matter how long you run it.

The situation gets worse in both directions simultaneously. As a light gets smaller, hits become rarer *and* — since a small light must be brighter to deliver the same illumination — each hit returns a larger value. Variance is the product of these, so it grows faster than either.

Which is unfortunate, because small bright lights are what lighting artists use: they produce crisp shadows and controlled falloff. The naive algorithm is worst exactly where the demand is.

2 Next-event estimation

Rather than hoping a path finds a light, sample one deliberately. At every surface vertex, choose a point on a light source, trace a shadow ray to it, and add its contribution using the **area form** of the rendering equation from Module 02 — with the geometry term converting the light's area measure into solid angle.

The path then continues as before by sampling the BSDF, but **emission is no longer counted when the continuation happens to strike a light**, because that contribution was already accounted for by the explicit light sample at the previous vertex. Counting both would double it.

	Before NEE	After NEE
Point lights	Impossible. Probability exactly zero.	Work correctly.
Small area lights	Thousands of samples to converge.	A handful.
Cost per vertex	One BSDF sample.	One BSDF sample plus one shadow ray .
Rays cast	Mostly continuation rays.	Mostly shadow rays — which is why Module 03's dedicated occluded() path is now worth its 2–3×.

AND NOW GLOSSY SURFACES BREAK

Consider a near-mirror surface. Its BSDF lobe is extremely narrow: only directions within a degree or two of the mirror direction have any significant value.

The sampled light point is almost never in that lobe. The BSDF evaluates to essentially zero, so the explicit light sample contributes nothing — at the cost of a shadow ray, every time.

Meanwhile, the BSDF-sampled continuation lands precisely in the lobe and may well strike the light — and we have just instructed it not to count emission.

So NEE alone is strictly worse than naive path tracing for sharp reflections of light sources. Glossy highlights vanish entirely. This is precisely the complementary failure identified in Module 02: two strategies, each excellent exactly where the other collapses, and the obvious combination — use both and add — double-counts every direction both can reach.

3 Multiple importance sampling

Veach and Guibas, 1995. The solution combines several sampling strategies with per-sample weights, and it is both simple and provably close to optimal.

$$F = \sum_s (1/n_s) \sum_i w_s(x_{s,i}) f(x_{s,i}) / p_s(x_{s,i})$$

where s indexes the strategies, n_s is the number of samples taken with strategy s , and w_s is a weighting function. **The estimator is unbiased for any weights satisfying a single condition:**

$$\sum_s w_s(x) = 1 \quad \text{for every } x \text{ with } f(x) \neq 0$$

That is the whole requirement. Any set of weights summing to one gives a correct answer; the choice affects variance only — the same structure as the choice of p in Module 05.

$$w_s(x) = n_s p_s(x) / \sum_t n_t p_t(x) \quad (\text{balance heuristic})$$

WHAT THE WEIGHTS ARE DOING

Each strategy's weight at a point is **its share of the total probability density** that the strategies collectively assign there. The shares sum to one by construction, so a direction both strategies can reach is counted exactly once.

The split is adaptive, and nobody has to classify the scene. Near a small light, the light-sampling density is enormous and the BSDF density is small, so light sampling receives nearly all the weight. Inside a tight specular lobe, the reverse holds.

The densities themselves carry the information about which strategy is appropriate, which is why MIS needs no heuristics about light size or material roughness and does not fail on configurations nobody anticipated.

Veach proved the balance heuristic is within a small additive constant of the variance of the best possible weighting — so there is very little left on the table.

The **power heuristic** raises each density to a power ($\beta = 2$ in practice) before normalising. It suppresses low-weight samples more aggressively and performs slightly better than the balance heuristic in most scenes. It is what production renderers use.

```

inline float mis_weight(float pdf_a, float pdf_b) {
    float a = pdf_a * pdf_a, b = pdf_b * pdf_b;    // power heuristic, beta=2
    return a / (a + b);
}

// 1. Light sample, weighted by how likely the BSDF was to find it.
auto [wi_L, Li, pdf_L] = scene.sample_light(h.p, rng.next2D());
if (pdf_L > 0 && !scene.occluded(h.p, wi_L)) {
    float pdf_B = h.bsdf->pdf(-r.d, wi_L);          // query, not sample
    float w      = mis_weight(pdf_L, pdf_B);
    L += throughput * w * h.bsdf->f(-r.d, wi_L)
        * fabs(dot(wi_L, h.ns)) * Li / pdf_L;
}

// 2. BSDF sample; if the continuation hits a light, weight symmetrically.
if (next.emits()) {
    float pdf_L2 = specular ? 0.0f : scene.pdf_light(h.p, wi_B);
    float w      = mis_weight(pdf_B, pdf_L2);
    L += throughput * w * next.emission();
}

```

TWO IMPLEMENTATION DETAILS THAT CAUSE MOST MIS BUGS

The other strategy's PDF is queried, not sampled. When evaluating the light sample's weight, you need the density the *BSDF* would have assigned to that direction — you do not draw another *BSDF* sample. This is exactly why Module 07 insisted that `pdf()` be callable independently of `sample()`.

Specular lobes set the other strategy's PDF to zero. A delta *BSDF* has infinite density in one direction, so its MIS weight is 1 and light sampling's is 0 — light sampling could never have produced that direction. Implement it by passing 0 for the other PDF, which makes the weight exactly 1.

Omitting the specular case is the single most common MIS bug, and its symptom is distinctive: mirrors and polished metals lose the reflections of light sources while everything else looks correct. If a chrome sphere shows the room but not the lamp, this is why.

Configuration	BSDF sampling	Light sampling	MIS
Small light, diffuse surface	Terrible	Excellent	Excellent
Large light, glossy surface	Excellent	Terrible	Excellent
Medium light, medium roughness	Poor	Poor	Good — better than either
Specular reflection of a light	The only option	Contributes zero	Correct, weight 1

The third row is the real argument for MIS. It is easy to assume MIS merely selects whichever strategy is better. It does more than that: in intermediate configurations, where *neither* strategy is adequate alone, the weighted combination has lower variance than either. That is the behaviour Veach's figure in the original paper demonstrates, and reproducing that figure is part of Project 1.

4 Debugging a path tracer

Symptom	Likely cause	Check
Image roughly twice as bright in directly lit areas.	Double counting. MIS weights missing, or emission counted after an NEE sample.	Disable NEE; if brightness becomes correct, the weighting is wrong.
Glossy and specular highlights of lights missing.	The specular MIS case is unhandled.	Check that a delta lobe forces the other PDF to zero.
Black image with point or very small lights.	No next-event estimation.	§2.
Dark speckle across all surfaces.	Self-intersection.	Module 03 §3 — use the ULP offset.
Isolated bright pixels on silhouettes and creases.	Intersection is not watertight.	Module 03 §2.
Uniformly too dark, worsening with roughness.	Microfacet single-scattering energy loss.	Module 07 — furnace test across the roughness range.
Image brightens the longer it runs.	A BRDF that creates energy.	Furnace test every material; check for albedo > 1 .
Converges cleanly to the wrong answer.	Bias. A clamp, a bounce limit, or a biased estimator.	Convergence plot (Module 05) — the curve flattens.

VALIDATE ONE COMPONENT AT A TIME

Direct lighting, diffuse only, one area light. There are configurations with analytic solutions — a disk light above a diffuse plane, for instance. Match them exactly. If this is wrong, nothing downstream can be right.

Add indirect. Render a Cornell box and difference it against PBRT's output. **The difference image should be unstructured noise.** Any visible structure — a bright wall, a dark corner — localises the bug immediately.

Add BSDFs one at a time, furnace-testing each before it enters a scene (Module 07 §4).

Finally enable MIS, and confirm the converged image does not change while the variance falls. **This is the critical test:** MIS is a variance reduction technique and must not alter the answer. If the converged image shifts when MIS is enabled, the weights do not sum to one and the estimator is biased.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
PBRT 4th ed. — Chapter 13, Light Transport I	Path tracing, NEE, and MIS with complete code. §13.10 on MIS is the implementation to compare yours against.
Veach & Guibas — Optimally Combining Sampling Techniques (1995, free)	The original MIS paper. Short, and the figure comparing strategies across light sizes is the one to reproduce.
Veach thesis — Chapter 9	The fuller treatment, including the optimality proof for the balance heuristic.
Benedikt Bitterli — rendering resources	Scenes in PBRT format with reference images. Your validation set for §4.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement the naive path tracer. Render a scene with a large area light and confirm it converges. Then shrink the light by factors of ten and plot variance against light solid angle.
- 2 Add a point light and confirm the naive tracer renders it black. This is worth seeing once.
- 3 Implement next-event estimation. Confirm point lights work and report the variance reduction on the small-light scene from exercise 1.
- 4 Construct a scene with a polished metal plane reflecting a small light. Render with NEE only and confirm the highlight is absent.
- 5 Implement MIS with the balance heuristic. Confirm the converged image matches the NEE-only result in diffuse regions and recovers the highlight.
- 6 Implement the power heuristic and compare variance against the balance heuristic on three scenes. Report whether the difference is worth having.
- 7 **Reproduce Veach's figure:** a row of plates of increasing roughness lit by lights of increasing size, rendered with BSDF sampling only, light sampling only, and MIS. This single image demonstrates the whole module.
- 8 Deliberately omit the specular MIS case and render a chrome sphere beside a light. Document the artefact.
- 9 Produce a four-way convergence plot — naive, NEE, BSDF, MIS — on the same scene. The MIS curve must sit below the others everywhere, and you must be able to explain each gap.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Write the naive path tracer from memory and say why it is unbiased.
- 2 Why does it fail for small lights, and why does the failure compound?

- 3 What does next-event estimation do, and why must emission then be skipped on the continuation?
- 4 Why does NEE alone fail on glossy surfaces?
- 5 State the MIS estimator and the single condition required for it to be unbiased.
- 6 What does the balance heuristic weight represent, and why does it adapt without being told the scene type?
- 7 Why is the other strategy's PDF queried rather than sampled?
- 8 How are specular lobes handled in MIS, and what is the symptom of getting it wrong?
- 9 Give four path tracer symptoms and their likely causes.
- 10 What must be true of the converged image when MIS is enabled, and why?